



COMPUTATIONAL ENGINEERING INTERNSHIP BROCHURE 2026-2027



TABLE OF CONTENTS

- 01 About IIT Hyderabad
- 02 Computational Engineering at IITH
- 03 Why Recruit Us?
- 04 Curriculum
- 05 Core Skills
- 06 Additional Skills
- 07 Projects
- 08 Past Recruiters
- 09 Previous year Stats
- 10 Previous year Internships
- 11 Contacts Us





Projects
4630+

Patents
530+

Startups
260+

Publications
11800+

- Established in 2008, IITH is one of India's newest IITs, rapidly gaining prominence in science and technology.
- The institute offers a strong interdisciplinary academic ecosystem spanning engineering, sciences, design, research, and entrepreneurship.
- Flexible academic structure with opportunities for minors, double majors, faculty-guided research, and hands-on learning.
- Strong research and innovation ecosystem supported by advanced laboratories, interdisciplinary research centers, and industry-driven projects.
- Active support for publications, patents, startups, and entrepreneurial initiatives through dedicated innovation and incubation programs.
- Vibrant student ecosystem promoting innovation, technical competitions, leadership, teamwork, and entrepreneurial thinking.



Computational Engineering at IIT Hyderabad equips students with modern computational, analytical, and problem-solving skills to address complex engineering and scientific challenges using advanced computing techniques.



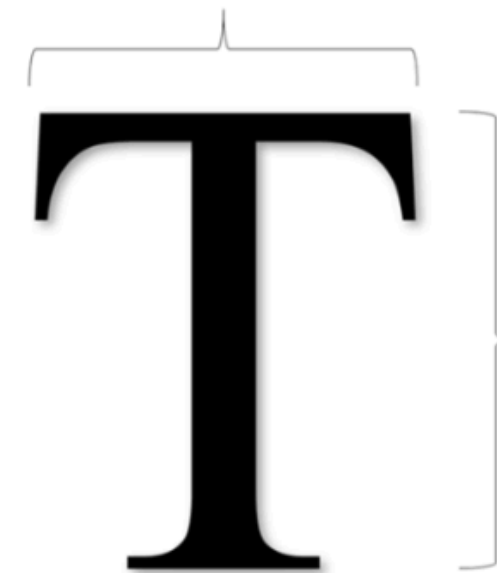
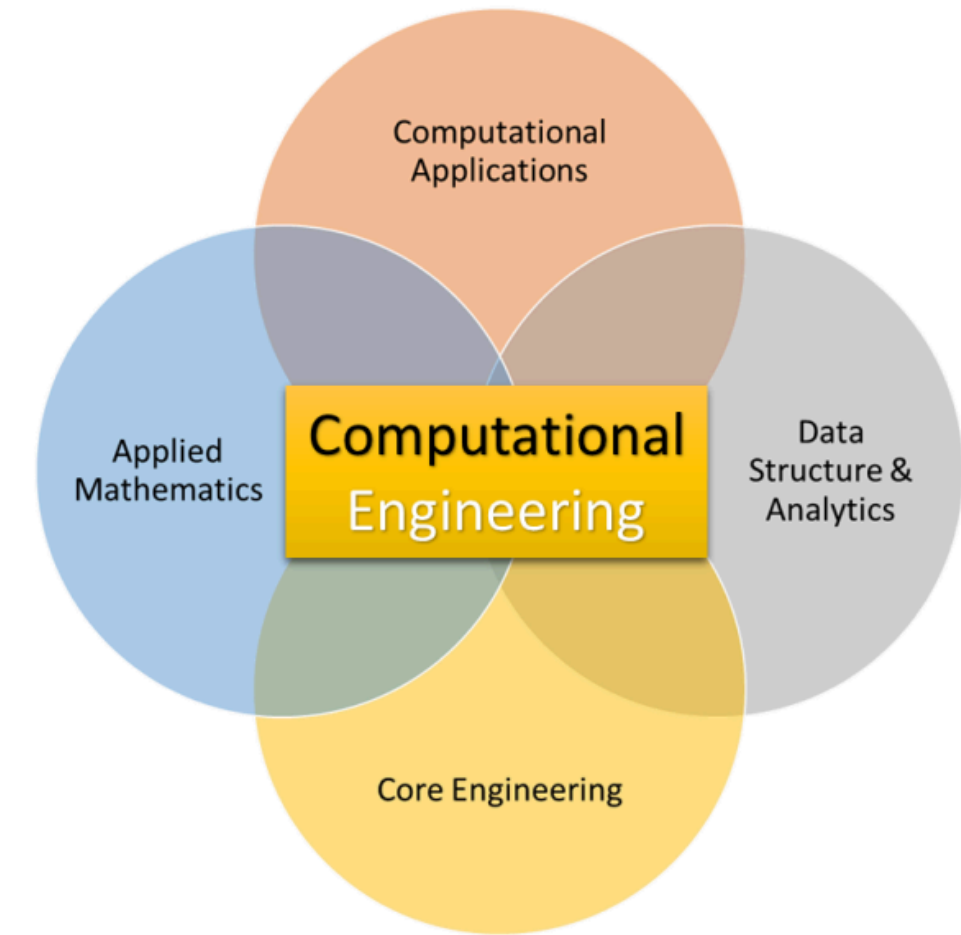
Students gain interdisciplinary exposure that enables them to solve real-world problems across multiple engineering and scientific fields.



Faculty members actively collaborate with academic and industry partners, imparting industry-relevant knowledge and ensuring that students are well-prepared for professional challenges.



With the growing demand for simulation, optimization, and data-driven engineering, Computational Engineering prepares students for emerging technology-focused and research-driven careers.



Computational Methods		
Scientific Computing	Data Structure & Analytics	Computational Applications

WHY RECRUIT US?

- IITH is consistently ranking among the top 10 engineering institutions in India by NIRF for its academic excellence.
- COE offers a strong curriculum which combines programming, numerical methods, data analysis, scientific computing, simulation, and engineering problem-solving, while developing core computational skills, algorithmic thinking, modelling and optimization along with the flexibility of electives for domain specialization.
- With a balanced mix of theory, laboratories, projects, and internships, students develop strong practical, technical, and problem-solving abilities for diverse roles across engineering and technology domains.

NIRF Overall 12	NIRF Engineering 7	NIRF Innovation 6
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03

YEAR	OPENING RANK	CLOSING RANK
2021	1171	1383
2022	1212	1633
2023	1304	1884
2024	1046	1790
2025	1368	1675

JEE Advanced Cutoffs of COE Branch

CURRICULUM

04

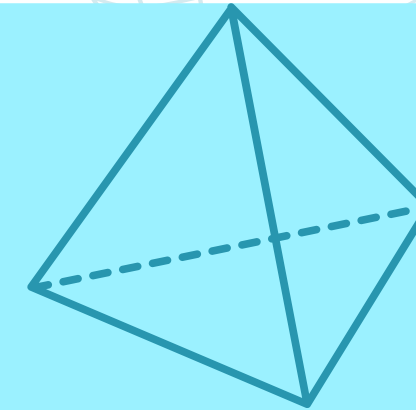


SOFTWARE

- Data structures and applications
- Concurrency control in Transactional Systems
- Operating Systems
- Parallel and Concurrent Programming
- Distributed Computing

- Computer Aided Numerical Methods
- Parallel Scientific Computing
- Computational Methods in Material Science
- Optimization Techniques

SCIENTIFIC COMPUTING

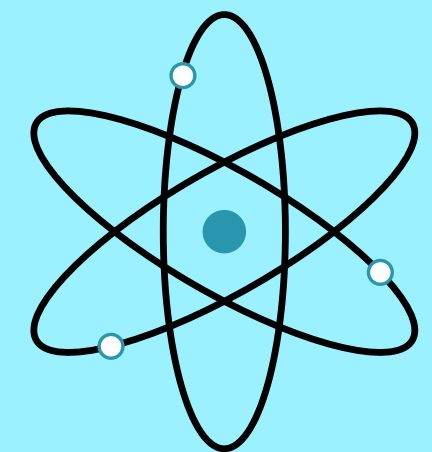


MATH AND ENGINEERING

- Calculus
- Linear Algebra
- Differential Equations
- Probability and Statistics
- Heat, Mass Transfer
- Modern Physics and Mechanics
- Thermodynamics
- Bioengineering and databases

- Finite Element Methods
- Computational Fluid Dynamics
- Digital Fabrication
- Atomistic Modeling
- Intro to Hardware Description Languages
- Mesoscale Modelling

MODELING AND SIMULATIONS



CORE SKILLS

- Solid grasp of data structures and computational problem-solving, enabling students to think critically and solve real-world engineering challenges efficiently.
- Understanding of how operating systems handle processes, memory, files, and concurrency help in making systems and backend development.
- Hands-on exposure to modelling, simulation, and optimization techniques for solving complex engineering and scientific problems.
- Practical experience in electronic structure and atomistic modelling, including Linux command-line workflows, DFT calculations, and Molecular Dynamics simulations.
- Strong understanding of computer-aided numerical methods including numerical integration and differentiation, direct & iterative solvers for linear systems, eigenvalue problems, and PDE schemes.
- Strong foundation in continuum mechanics and fluid flow principles; ability to apply FEM for structural analysis and stress-strain simulations; hands-on exposure to CFD techniques for modeling, analyzing, and optimizing behaviors in engineering systems.
- Conceptual knowledge of Heat, Mass Transfer and thermodynamic principles governing engineering systems.
- Hands-on experience in sourcing biological data, managing datasets, and using visualization them to analyze molecular structures and interactions.

ADDITIONAL SKILLS

Students at IITH actively explore technologies beyond the classroom through projects hackathons, and research

- **C/C++**

A strong foundation in data structures, algorithms, operating systems, and OOP

- **App & Web Development**

HTML, CSS, JavaScript, Flutter, Node.js, Next.js, MongoDB, and Firebase

- **AI/ML**

PyTorch, TensorFlow, and Scikit-learn

- **Verilog / VHDL**

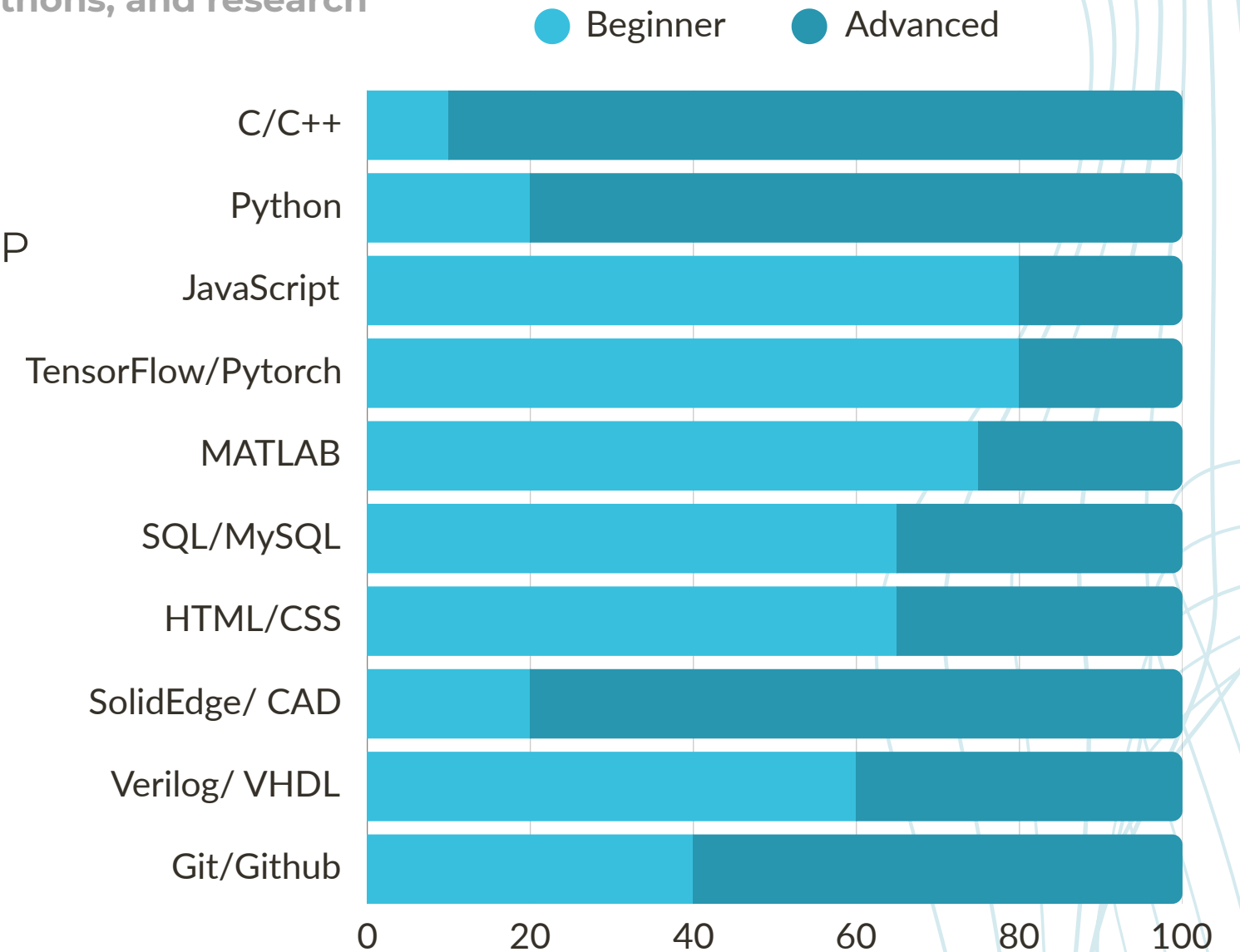
Design, simulate and implement digital circuits and hardware systems.

- **SolidEdge / CAD / Blender**

3D modeling, product design, and Architectural Design.

- **Game Development**

C# and Unity engine



Student Clubs that support this learning

- Epoch-AI/ML

- Glitch-Game dev

- Lambda- Web dev

- Kludge-CyberSecurity

- Inero- Competitive programming

- BlockSpace- BlockChain

PROJECTS

01

Molecular Dynamics Engine

Solid-liquid phase transitions using Lennard-Jones potential & Velocity Verlet integration (Python)

02

Key Management Software

Secure full-stack application using React, Go & PostgreSQL

03

OS Performance Improver (RL)

Reinforcement Learning to optimize Linux OS performance (C/C++/Python)

04

ODE/PDE Solver

Solver implemented using numerical methods including RK2, RK4, ABM2, ABM4, FDM, and the Shooting Method

05

xv6 OS Extensions

Extended the xv6 OS with custom schedulers (MLFQ), virtual memory management, system calls, and RAID storage. (C)

06

ML Data Annotation Pipelines

ML Data Annotation Pipelines using semi-supervised learning, CVAT, and Python.

07

MacOS Process Monitor

Real-time system & resource monitoring using C++, Objective-C++, CMake

08

Portfolio Optimiser

Optimizes portfolio of Indian stocks using Modern Portfolio Theory, Streamlit, Scipy & Plotly

09

TradePro Stock Forecasting

A stock forecasting platform made using ARIMA, Prophet, LSTM & Random Forest models.

PAST RECRUITERS

08



THOMSON REUTERS®



Microsoft



Qualcomm



D E Shaw & Co



Sagri

SIEMENS

Goldman Sachs



BOSCH



CRYSTAL BALL



Google Summer of Code



Accelquant



teradata.

PREVIOUS YEAR STATS

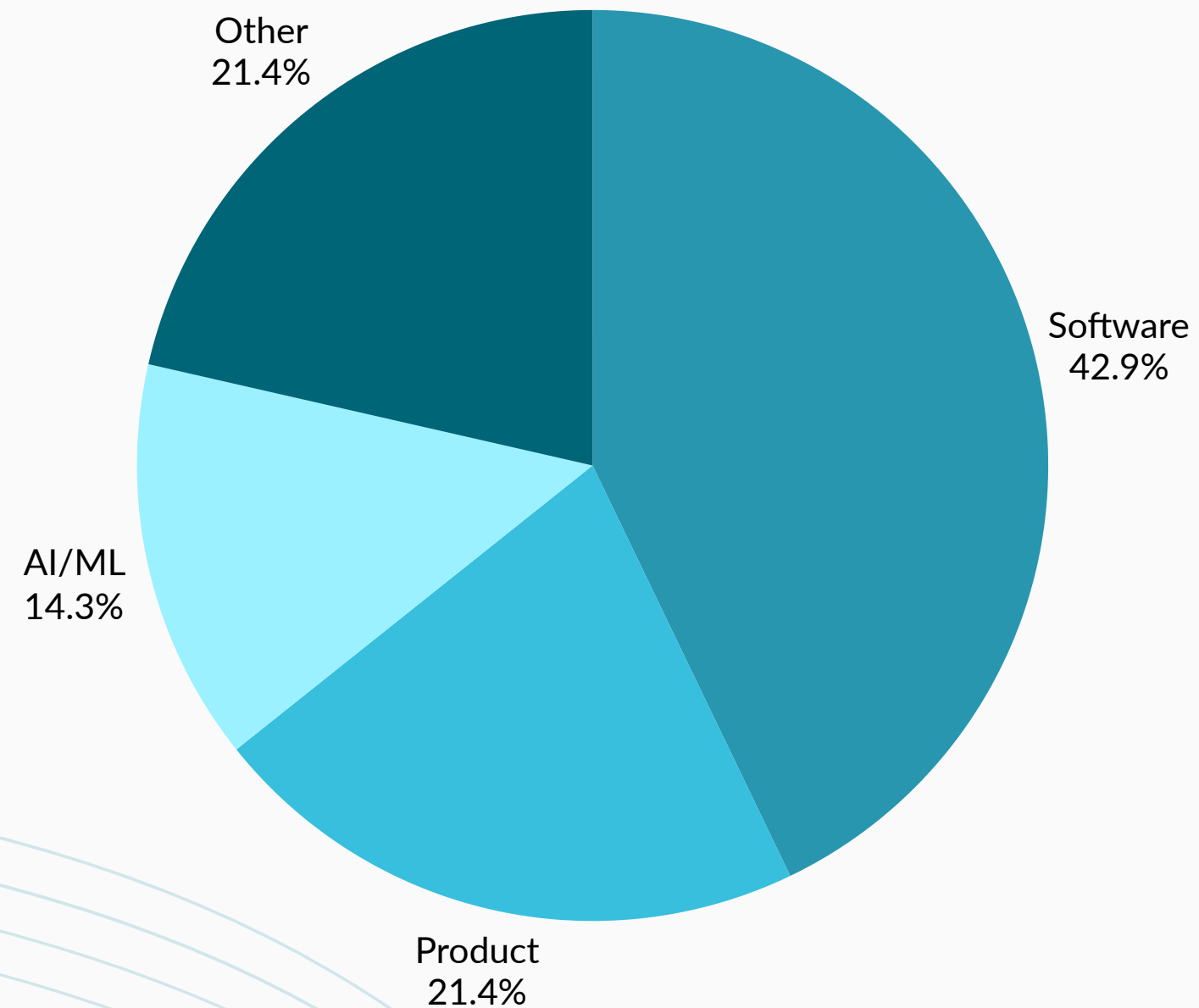
09

100%

Placement Percentage

100+

Companies offered
Internship



PREVIOUS YEAR INTERNSHIPS

10

Company	Role
Microsoft	SWE Intern
Qualcomm	Software Engineer
Expedia Group India	Mobile Engineer
Devrev	SDE Intern
Verint	Software Intern
Murata Machinery	Research and Development Intern

Company	Role
Teradata	Data, AI/ML Intern
Oracle	Applications Engineer
Goldman Sachs	Quant
Morgan Stanley	Applications Development Intern
Samsung	Software Engineer
Thomson Reuters	Applied AI Research Intern

CONTACT US



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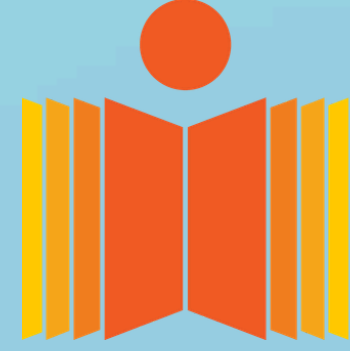
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భారతీయ సాంకేతిక విజ్ఞాన సంస్థ హైదరాబాద్
भारतीय प्रौद्योगिकी संस्थान हैदराबाद
Indian Institute of Technology Hyderabad

THANK YOU